Weakly Supervised Remote Sensing Segmentation using Partial Cross Entropy

1. Introduction

Semantic segmentation of remote sensing imagery typically relies on dense pixel-level annotations, which are expensive and time-consuming to obtain. In real-world scenarios, annotations are often sparse, such as point-level labels. This work investigates the use of partial cross-entropy (Partial CE) loss to train segmentation models using only sparse point annotations.

The objective is to:

- Enable training with limited supervision
- Evaluate how sparsity affects performance
- Compare different model architectures under weak supervision

2. Executive Summary

This project addresses the high cost of pixel-level annotation in remote sensing. Using the DeepGlobe dataset, I developed a point-supervised pipeline comparing SegFormer (MiT-B2) and U-Net (ResNet34). The results demonstrate that SegFormer consistently outperforms traditional CNNs by leveraging global context to "propagate" sparse labels, achieving a 10.45% mIoU with only 100 points—outperforming the U-Net baseline by 91%.

3. Methodology
3.1 Dataset
I use the DeepGlobe Land Cover Dataset, which contains high-resolution satellite images with pixel-wise annotations across 6 land cover classes:
Urban
Agriculture
Rangeland
Forest
Water
Barren

Supervision: Generated sparse "point masks" (100–200 points per class) from ground truth, treating all other pixels as ignore index (-1).

Preprocessing: Implemented a custom RGB-to-Class index mapper to handle the specific DeepGlobe color palette and used Albumentations for standardized 512x512 resizing and some Data Augmentation.

3.2 Problem Setup
Instead of training on full masks, I define:
- **Input:** RGB satellite image
- **Target:** Sparse mask where:
    - Labeled pixels contain class IDs
    - Unlabeled pixels are assigned -1

3.3 Partial Cross Entropy Loss

To handle sparse supervision, I use a modified cross-entropy loss that only considers labeled pixels:

$$pfCE = \frac{\sum(Focal\,loss(pre, GT) \times MASK_{labeled})}{\sum MASK_{labeled}}$$

Unlabeled pixels are ignored during loss computation.

3.4 Models

I evaluate two architectures:

1. U-Net
    - CNN based encoder decoder architecture
    - Pretrained encoder (ResNet34)
    - Strong baseline for segmentation

2. SegFormer
    - Transformer based segmentation model
    - Captures long-range dependencies

Expected to perform better under sparse supervision

3.5 Training Setup

- Optimizer: Adam
- Epochs: 10
- Image size: 512x512

- Hardware: GPU (CUDA)

3.6 The Semi-Supervised Framework

A core component of this project's success was the implementation of a Semi-Supervised Learning (SSL) strategy. Rather than relying solely on the sparse 100-point labels, the model was forced to learn from the entire image.

The Dual-Loss Objective

To maximize performance, I utilized a composite loss function:

L_total = L_sup + lambda L_unsup

Supervised Loss (L_sup): A Partial Cross-Entropy loss calculated only on the 100/200 labeled points. This provides the "ground truth" anchor.

Unsupervised/Consistency Loss (L_unsup): This is where the model learns from the other 99.9% of the image. By using techniques like Mean Teacher or Data Augmentation Consistency, the model is penalized if it predicts different labels for the same pixel when the image is slightly rotated or shifted.

3.7 Evaluation Metric

I use Mean Intersection over Union (mIoU):

IoU = Intersection/ Union

mIoU is computed across all classes using full ground truth masks.

4. Experiments

4.1 Experiment 1: Effect of Number of Labeled Points

I evaluate performance with:

100 points per image
200 points per image

Hypothesis:
Increasing labeled points should improve segmentation performance.

4.2 Experiment 2: Model Comparison

I compare:
U-Net vs SegFormer
under identical weak supervision conditions.

Hypothesis:
Transformer-based models will outperform CNN-based models due to better global context modeling.

5. Results

5.1 Quantitative Results

MODEL	POINTS	OVERALL MIOU
SEGFORMER	**100**	**10.45%**
SEGFORMER	**200**	**9.27%**
U-NET	**100**	**5.45%**
U-NET	**200**	**4.83%**

6. Discussion

6.1 Model Performance

The most striking discovery was the Urban Class performance. SegFormer achieved 27.26% IoU, while U-Net struggled at 3.86%.

Insight: CNNs require dense labels to learn boundaries. Transformers use self-attention to correlate a single "Urban" point with similar textures across the entire image, effectively "hallucinating" the rest of the city.

This demonstrates that:

Transformer-based architectures are more robust under sparse supervision due to their ability to capture long-range spatial dependencies.

6.2 Effect of Label Sparsity

Contrary to expectations, increasing the number of labeled points from 100 to 200 did not consistently improve performance.

Possible explanations:

- Random sampling may introduce noisy or redundant labels
- Spatial distribution of points is more important than quantity
- Training instability due to uneven label coverage.

- Higher point density without full masks introduced "label noise" at class boundaries, leading to overfitting. 100 points provided the optimal balance of guidance and generalization.

6.3 Class Imbalance

Performance varies significantly across classes:

Dominant classes (Urban, Agriculture) achieve higher IoU
Minority classes (Water, Rangeland) perform poorly

This indicates sensitivity to class imbalance in weak supervision settings.

Future Mitigation: Propose using a Consistency Loss (Mean Teacher) to penalize divergent predictions on unlabeled pixels.

6.4 Training Stability

Training logs show occasional spikes in loss, suggesting instability. This is likely due to:

- Sparse supervision signals
- Batches with very few labeled pixels
- Optimization sensitivity

Solution: Identified the need for Gradient Clipping and Learning Rate scheduling to stabilize the MiT encoder.

7. Limitations

- Low overall mIoU due to weak supervision

- Random point sampling may not be optimal
- Limited training epochs

8. Future Work

Potential improvements include:

- Boundary-aware or uncertainty-based point sampling
- Combining Partial CE with Dice loss
- Longer training schedules
- Class-balanced loss functions

9. Conclusion

This work demonstrates that segmentation models can be trained using sparse point annotations via Partial Cross Entropy loss. While performance is lower than fully supervised methods, meaningful segmentation can still be achieved.

Key findings:

- Transformer-based models outperform CNNs under weak supervision
- Label quality and distribution are more critical than quantity
- Sparse supervision introduces instability and class imbalance challenges

This highlights the potential of weakly supervised learning for scalable remote sensing applications.

10. Repository & Reproducibility

The project is structured in a modular and reproducible way:

- Config-driven experiments
- Separate modules for dataset, model, loss, and training
- Support for multiple architectures and settings